



MINING DEALER BEST PRACTICE

Machine Energy Maps

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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1.0 Introduction

Mining equipment is fundamental in achieving production goals at any mining operation. To ensure the availability and reliability of the equipment, it is necessary to include maintenance and repair tasks that must be executed with efficiency, quality, and safety.

When conducting maintenance procedures, blocking or deactivating stored energy in equipment is a routine task. However, this requires rigorous follow-up of established safety procedures - procedures known in Finning as "The Rules of Life". The Rules of Life are a set of rules that define behaviors of mandatory compliance for all employees, disobedience of these rules is considered unacceptable.

We have developed this Best Practice to ensure that all personnel involved in the maintenance and repair processes identify these life rules when working with mining equipment. Elimination of energy points in a practical and fast way will directly improve efficiency and increase safety standards.

2.0 Best Practice Description

This best practice consists of the development and implementation graphics for the different Caterpillar equipment models, with a visual and easy understanding representation of the energy points that must be neutralized or blocked before carrying out the maintenance and repair work.

This best practice (BP) complements the information of the maintenance guidelines and procedures and increases access to information – especially for technicians that are in the training process – by summarizing and facilitating the identification of the components and systems of the equipment.

2.1 Opportunity

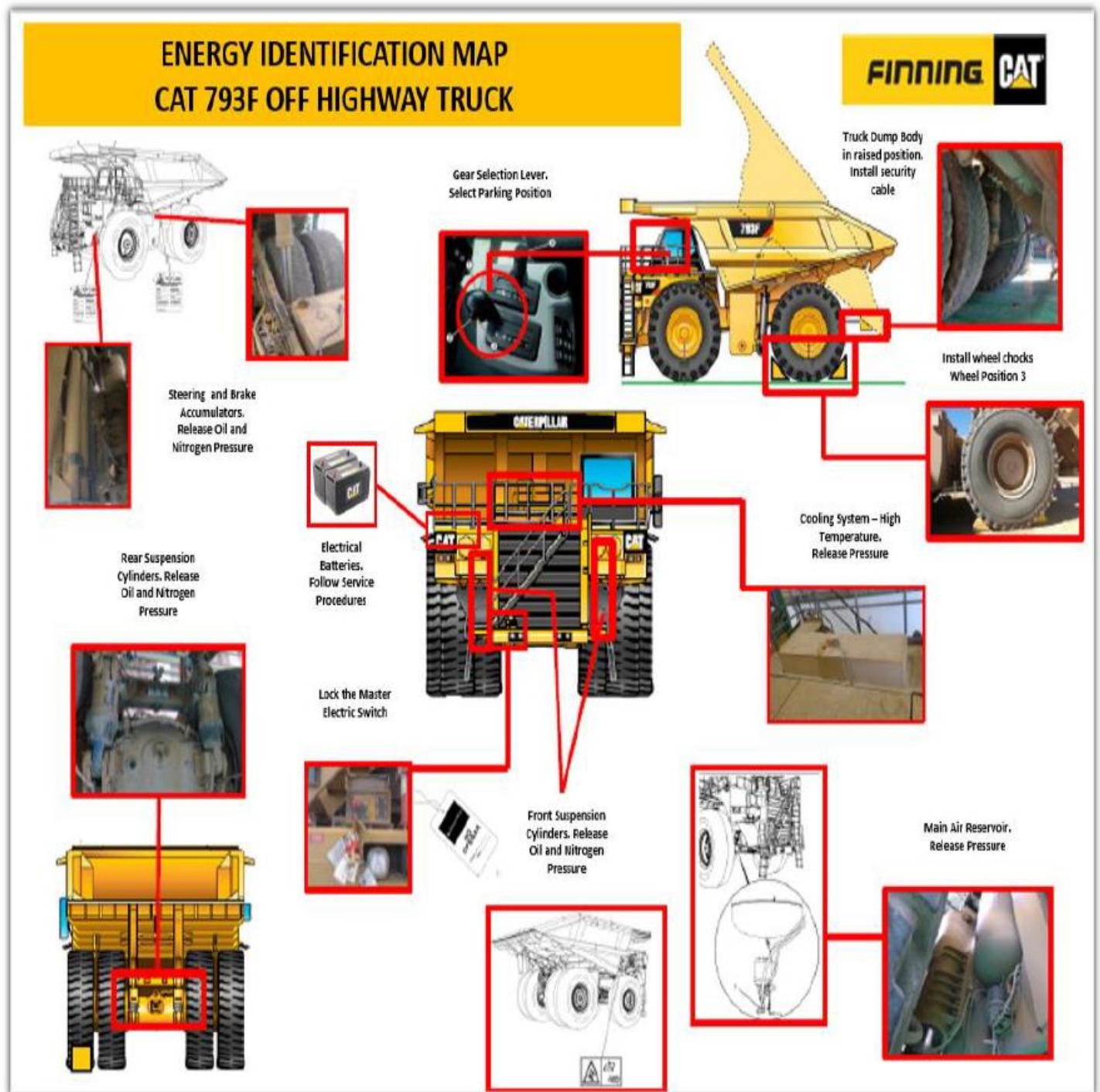
In 2016, a serious accident involving energy exchange occurred due to the release of oil subjected to residual pressure. This accident brought attention to effective procedures and controls of energy blockage/elimination.

First, measures were aimed at identification of energy only in areas and places where measuring instruments were used. Once all APCIEs (High Potentials for Energy Exchange) were identified, the need to continue to strengthen and develop the content was identified and information was extended to all the machine systems. Communication of this information to the end users was simple, graphical, didactic and in a quick-access format.

The first equipment models selected for the development of the "Equipment Energy Maps" were the CAT-793F Truck and the CAT 994H Wheel Loader. These diagrams identify each of the components and systems that have or may have energy and include basic energy isolation procedures (system or component to block and / or isolate - Blocking point - How? - With what? - Risk).

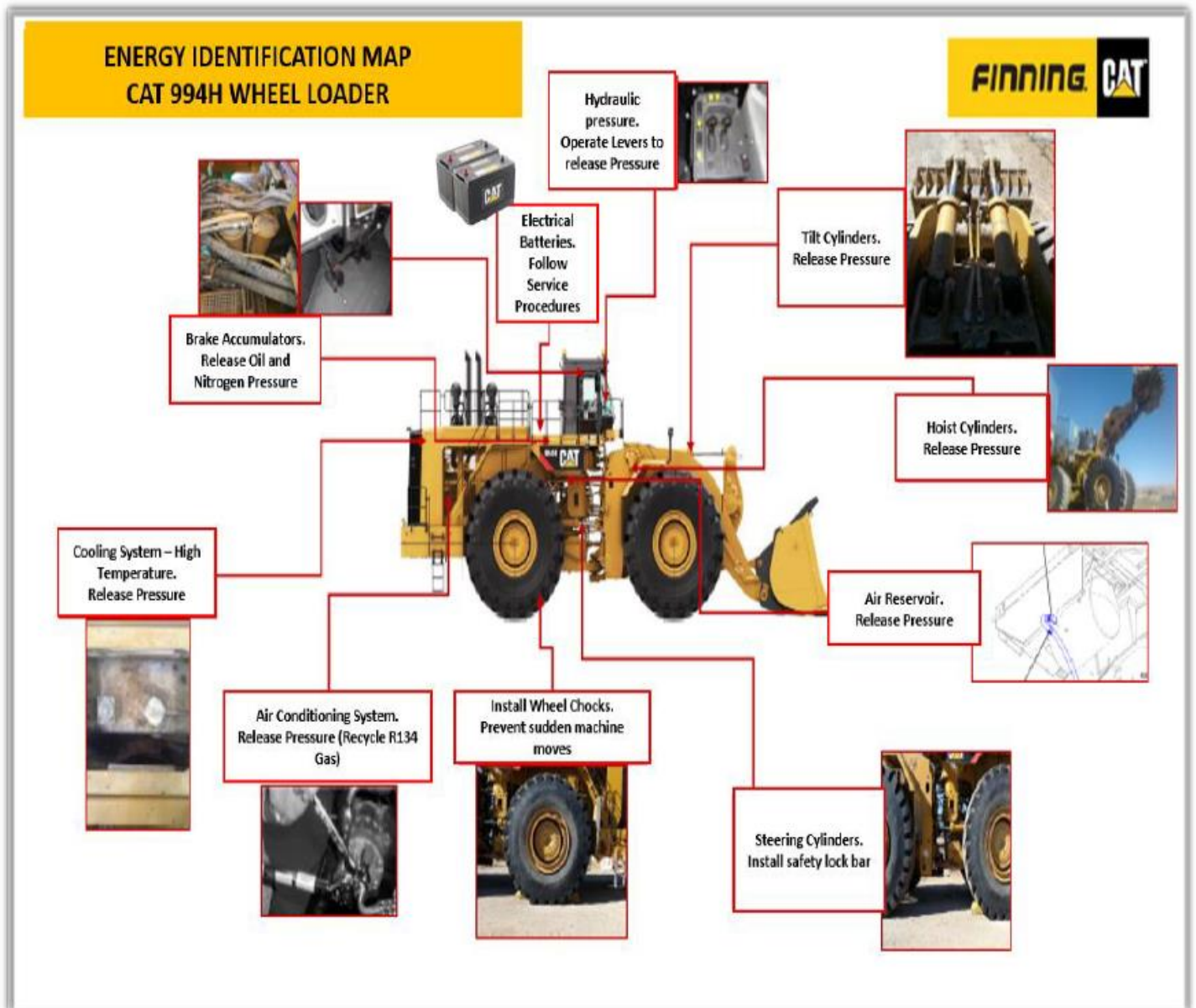
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These diagrams were installed at key points of the workshops, using giant graphics to ensure adequate visualization of them by the technicians. Below are two examples of the developed Equipment Energy Map:



Energy Map for CAT 793F Truck

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Energy Map for CAT 994H Wheel Loader

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3.0 Implementation Steps

We recommend the following steps when implementing this Best Practice

3.1 Tasks assignment

Once the need to generate energy maps is identified, it is necessary to bring together the areas of the organization involved in the development and implementation of this BP (In our case: SHEQ, Engineering, Operations, Mentoring).

Each of these areas must assume the responsibility of developing certain tasks – according to their specialization – forming an effective project team. The area of prevention and operations are those who define what is required to improve those identified areas of high potentials for energy exchange. In this case, the Engineering and Mentoring Area, due to their technical capabilities, were responsible for identifying and delivering the complete and detailed information of each machine type, consolidating the necessary data, and ensuring that no additional information was required.

3.2 Select Format

Once the Engineering and Mentoring area compile the technical information of the equipment and the potential energies, they proceed to choose the design and format of the maps. In our situation, the Safety area (SHEQ) was responsible for the execution of this task, since they are the ones with experience to define the visual requirements, so that the message gets communicated effectively to the service personnel (See the examples of Energy Maps).

3.3 Maps Distribution

Once the first draft of the Energy Maps has been completed, we recommend distributing and analyzing them with the technicians. At this stage, make sure to gather suggestions from the technicians and define the final form of implementation for the field service and the workshop. For the field service, we recommend including the Energy Maps information in the service folders with instructions to be applied to scheduled and unscheduled jobs.

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3.4 Printing and Implementation

The printing of the Energy Maps in a big format, was done locally with the following provider:

Artegraf E.I.R.L
 Rut 76056383-8
 Av. Argentina N°1312, Antofagasta, Chile
 Contacto Rossana Acuña, artegraf@vtr.net

The Energy Maps were installed in the areas of the workshop corresponding to Preventive Maintenance and execution of Scheduled and Non-Scheduled tasks.



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3.5 Start-up and Benefits Validation

Once the information / maps get implemented in the workshop, field operations, and shop work areas, their proper use should be monitored and their positive contribution in avoiding accidents validated.

4.0 Benefits

The main benefits of this best practice are in: Safety and Efficiency of the Technicians.

4.1 Safety

With the implementation of these maps, technicians were trained and made aware of the importance of blocking and eliminating stored or residual energies that represented a high risk in the execution of maintenance and repair tasks. Since the implementation of this Best Practice no incident has occurred due to this type of risks.

4.2 Efficiency

It should be noted that the Equipment Energy Maps, consolidate in a single document all the information required by the technicians, improving the information search times that is normally done using SIS (Service Information System) or Workshop service manuals. Improved times that positively impact the efficiency of maintenance and repair tasks.

5.0 Resources Required

Resources required to implement this Best Practice are as follows:

- MS Power Point Software
- Printing Cost (USD 90 each map approx.)

6.0 Supporting Attachments - References

N/A.

7.0 Related Best Practices

N/A.

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